

RAHUL SODANAPALLI

Front-End Engineer | React & TypeScript Specialist | Micro-Frontend Architect

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PROFESSIONAL SUMMARY

Results-driven Senior Front-End Engineer with 2+ years of experience delivering production-grade enterprise web applications for UAE government and utility clients. Expert in React.js, TypeScript, and Micro-Frontend Architecture (Module Federation / Vite). Proven track record of integrating AI/LLM features, building RBAC workflows, optimizing Core Web Vitals (sub-1.5s FCP), and leading shared component library development (80+ components). Experienced with Agile/Scrum, Azure DevOps CI/CD, and AWS S3 deployments.

TECHNICAL SKILLS

Languages JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL

Frameworks React.js (v18/v19), Next.js, Node.js, Express.js

Architecture Micro-Frontend (Module Federation, Vite Plugin Federation), SPA, REST APIs

UI Libraries Material UI (MUI), Ant Design, Bootstrap, Tailwind CSS v4, Framer Motion, GSAP

State Mgmt Redux Toolkit, RTK Query, Context API

Forms & Valid. React Hook Form, Zod, Yup

Data visualization ApexCharts, Recharts, Leaflet Maps

Backend/DB Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt, Nodemailer

Performance Code Splitting, Lazy Loading, Webpack, Vite, Core Web Vitals (FCP, CLS)

Dev & DevOps Git, GitHub, Azure DevOps, Jenkins, AWS S3, CI/CD Pipelines, Cursor AI IDE

Testing/QA Jest (familiar), Unit Testing, Code Reviews, WCAG Accessibility

Methodologies Agile, Scrum, AI-Assisted Development

WORK EXPERIENCE

Software Engineer — Front-End / Full-Stack | [Innovatech Technology Solutions](#) July 2024 – Present | Hyderabad, India

- Architected Micro-Frontend ecosystems for two major UAE government/enterprise platforms (DoE & TAQA) using Vite Plugin Federation (Module Federation), decoupling Gas Registry, Finance, Compliance, Economics, and Incident Reporting into independently deployable pipelines.
- Integrated Generative AI Document Validation (AiFileView widget) that triggers background LLM parsing on uploaded files (Emirates IDs, permits, certificates), surfacing confidence scores and validity checks in real-time.
- Engineered a real-time AI Chat Drawer (ChatDrawer) powered by Server-Sent Events (SSE) for regulator-to-document chat with optimized text-decoding algorithms for low-latency streaming.
- Designed and published a shared component library of 80+ reusable UI components (data tables, charts, loaders, buttons, hooks), guaranteeing visual consistency and reducing bundle size across all micro-frontends.
- Delivered 30% reduction in page load time via code splitting, lazy loading, skeleton loaders, and Webpack/Vite bundle tuning; achieved sub-1.5s First Contentful Paint (FCP) and minimized Cumulative Layout Shift (CLS).
- Implemented Granular RBAC with a custom GateKeeper component enforcing privilege-based module rendering and multi-level approval workflows (Engineer → Section Head → Director).
- Built a full-stack personal portfolio platform (React 19, Node.js, Express, MongoDB) featuring JWT auth (HTTP-Only cookies), bcrypt, CRUD admin dashboard, Nodemailer SMTP pipeline, and physics-based canvas particle animations with Lenis smooth-scroll.
- Developed Global Localization Layer supporting English and Arabic (RTL) views using custom Redux state provider and MUI/AntD theming.
- Built complex interactive data tables with Material React Table supporting server-side pagination, column pinning, and advanced filtering; integrated analytics dashboards using ApexCharts, Recharts, and Leaflet Maps.
- Configured multi-environment CI/CD pipelines (.env.localdev / ixqa / uat / prod) with Jenkins automation and automated deployments to AWS S3.
- Integrated 10+ RESTful APIs using Axios with custom Interceptors for authentication, dynamic token handling, and global error management via RTK Query.
- Conducted unit testing, peer code reviews, and maintained technical documentation across 3+ production releases in Agile/Scrum sprints.

KEY PROJECTS

Abu Dhabi Department of Energy (DoE) — Unified Regulatory & Licensing Portal — Frontend Developer

Mission-critical enterprise portal for UAE Government to regulate energy licenses, gas inspection registries, HSE campaigns, and financial audit tracking.

- Implemented Micro-Frontend Architecture with Vite Plugin Federation across Gas Registry, Finance, and Committee Management domains.
- Co-developed AI Document Validator widget with LLM-backed confidence scoring on Emirates IDs, permits, and compliance certificates.
- Built real-time AI Chat Drawer using Server-Sent Events (SSE) for in-document querying by regulators.
- Designed dynamic multi-step forms with document upload, multi-level approval flows, and real-time status tracking (responsive across all devices).
- Delivered WCAG-compliant modules with English/Arabic (RTL) localization using Redux-driven theme provider.

Tech Stack: React.js, TypeScript, MUI, Ant Design, Redux Toolkit, RTK Query, Axios, ApexCharts, Recharts, Leaflet Maps, Module Federation, Vite, SSE, RBAC

TAQA Water Solutions — Enterprise Micro-Frontend Platform (Shell/Host) — Frontend Developer

High-performance Micro-Frontend ecosystem orchestrating Compliance, Economics, ADESP, and IRS business modules with centralized design system.

- Architected the core host shell container (taqa-shell-ui) using Module Federation for independent sub-team deployments without system-wide rollbacks.
- Published shared design library of 80+ components (tables, charts, loaders, hooks) reducing application bundle size and ensuring visual consistency.
- Engineered GateKeeper RBAC component for privilege-based rendering and action-level feature gating.
- Configured global state (Redux Toolkit) and automated HTTP patterns with Axios Interceptors for auth, dynamic requests, and token expiration.
- Set up multi-environment Jenkins CI/CD pipelines with automated AWS S3 deployments across dev/QA/UAT/prod.

Tech Stack: React.js, TypeScript, Redux Toolkit, RTK Query, Ant Design, ApexCharts, Axios, Module Federation, Vite, Jenkins, AWS S3, Responsive Design

RahulBuilds.dev — Full-Stack Technical Portfolio Platform — Full Stack Developer

Production SPA showcasing advanced front-end engineering, secure full-stack architecture, and AI-powered development practices.

- Engineered physics-based mouse-attraction particle canvas animation (HTML5 Canvas) and magnetic micro-interactions using Framer Motion, GSAP, and Lenis smooth scrolling.
- Optimized Core Web Vitals via dynamic code-splitting and asset lazy-loading with skeleton loaders; achieved sub-1.5s FCP with minimal CLS.
- Built secure JWT admin CRUD dashboard with HTTP-Only cookies, bcrypt password hashing, and live content management (projects, skills, experiences).
- Implemented automated Nodemailer SMTP pipeline routing contact inquiries to email and logging records in MongoDB via Mongoose schemas.

Tech Stack: React 19, TypeScript, Node.js, Express, MongoDB, Mongoose, Redux Toolkit, RTK Query, Tailwind CSS v4, Framer Motion, GSAP, Lenis, JWT, bcrypt, Nodemailer, Zod

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science & Engineering Jul 2019 – Aug 2023

Presidency University, Bangalore, Karnataka, India

KEY ACHIEVEMENTS

- International Enterprise Delivery: Delivered production applications actively used by UAE government and utility stakeholders across Abu Dhabi regulatory sectors.
- AI Integration in Production: Integrated LLM-powered document validation and real-time AI streaming chat into live government portals, reducing manual support dependency.

- Micro-Frontend Pioneer: Independently architected and implemented Module Federation-based MFE ecosystems for two large-scale enterprise clients.
- Shared Library Author: Designed and shipped an 80+ component design system adopted across multiple business domains and sub-teams.
- Performance Champion: Consistently achieved 30%+ page load reductions and sub-1.5s FCP across all production modules.

LINKS

Portfolio: rahulsodanapalli.vercel.app

GitHub: github.com/rahulsodanapalli

LinkedIn: linkedin.com/in/rahul-sodanapalli-0a49062b3